

2.4 Normality and smoothness

Def: An irreducible affine variety V is called **normal** if its coordinate ring $\mathbb{C}[V]$ is normal. (An integral domain R is normal if it is integrally closed in its field of fractions K).

Def: An affine semigroup $S \subset M \cong \mathbb{Z}^d$ is **saturated** in M if $km \in S$ for $k \in \mathbb{N}_{>0}$, $m \in M$ implies that $m \in S$.

Thm: $Y = \text{Specm}(\mathbb{C}[S])$ affine toric variety of an affine semigroup $S \subset M$ such that $\mathbb{Z}S = M$. The following statements are equivalent:

1. Y is normal
2. S is saturated in M
3. $S = S_\sigma$ for some $\sigma \in N_{\mathbb{R}}$. ($S_\sigma = \sigma^\vee \cap M$)

Example: non-normal variety Y_A , $A = \begin{pmatrix} 2 & 3 \end{pmatrix}$
 $S = \mathbb{N}A = \{0, 2, 3, 4, \dots\}$ not saturated in $\mathbb{Z}$

Cor: The affine toric variety Y_A is normal if and only if the affine semigroup $\mathbb{N}A$ is saturated in $\mathbb{Z}A$. $S = \mathbb{N}A$, $M = \mathbb{Z}A$

Thm: The affine toric variety $Y_\sigma = \text{Specm}(\mathbb{C}[S_\sigma])$ is **smooth** if and only if σ is a smooth rational cone.
 Δ minimal generators are a subset of a $\mathbb{Z}$ -basis of N . Def 2.2.8

Prop: The cardinality $|H|$ of the Hilbert basis for S_σ is the smallest integer n for which Y_σ can be embedded in $\mathbb{A}^n$.
 $\mathcal{H} \subset S_\sigma$ be the set of all irreducible elements of S_σ .

Chapter 3 Polytopes and projective toric varieties.

In Chapter 2,

rational cone σ can generate an normal affine toric variety $Y_\sigma = \text{Specm}(\mathbb{C}[S_\sigma])$.
polytope P projective

3.1 convex lattice polytopes

Def: A (convex) polytope P in $M_{\mathbb{R}}$ is the convex hull of a finite set of points $A = \{a_1, \dots, a_n\}$ in $M_{\mathbb{R}}$:

$$P = \text{Conv}(A) = \left\{ \sum_{i=1}^n \lambda_i a_i \in M_{\mathbb{R}}, \lambda_i \in \mathbb{R}, \sum_{i=1}^n \lambda_i = 1, \lambda_i \geq 0 \right\} \subset M_{\mathbb{R}}.$$

If $A \subset M \subset M_{\mathbb{R}}$, then P is a lattice polytope.

$\dim P$ = dimension of the smallest affine subspace of $M_{\mathbb{R}}$ containing P .

Def: $u \in M_{\mathbb{R}} \setminus \{0\}$, $b \in \mathbb{R}$.

$$H_{u,b} = \{m \in M_{\mathbb{R}} : \langle u, m \rangle + b = 0\}$$

$$H_{u,b}^+ = \{m \in M_{\mathbb{R}} : \langle u, m \rangle + b \geq 0\}$$

$H_{u,b} \cap P$ is a face of P if $P \subset H_{u,b}^+$ and $b = -\min_{m \in P} \langle u, m \rangle$.

Def: simple: each vertex is contained in dimension-many facets. (codimension-1 face)

simplicial: faces are simplices. (A d -simplex is a polytope of dimension d with the minimal amount of vertices.)

Def: $\left\{ \begin{array}{ll} \text{vertex representation} & V\text{-representation} \\ \text{half-space representation} & H\text{-representation} \end{array} \right.$

$$P = \text{Conv} \{v \in M_{\mathbb{R}} : v \text{ is a vertex of } P\}$$

$$P = \bigcap_{\alpha \text{ facet of } P} H_{u_\alpha, b_\alpha}^+$$

Def: dilation of the polytope P : $\forall \lambda \in \mathbb{R}_{\geq 0}, \lambda \cdot P = \{\lambda p : p \in P\}$

Thm: Let $P \subset \mathbb{R}^n$ be a convex lattice polytope of dimension d . Ehrhart function $E_P : \mathbb{N} \rightarrow \mathbb{N} \quad k \mapsto |k \cdot P \cap \mathbb{Z}^n|$ is a polynomial with leading term $\text{Vol}(P) k^d$.

3.2 Affine charts of a projective toric variety

Def: standard affine charts of $\mathbb{P}^{n-1}$: $U_i = \mathbb{P}^{n-1} \setminus V(x_i) \quad i=1, \dots, n$.

$$\begin{aligned} \text{Isomorphism: } \varphi_i : U_i &\rightarrow \mathbb{C}^{n-1} \\ (x_1, \dots, x_n) &\mapsto \left(\frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right) \end{aligned} \quad U_i \cong \mathbb{C}^{n-1}$$

$$\bigcap_{i=1}^n U_i \cong (\mathbb{C}^*)^{n-1}$$

Standard affine covering $X_A = \bigcup_{i=1}^n (X_A \cap U_i)$

Prop: The affine chart $X_A \cap U_i$ of X_A is isomorphic to the affine toric variety X_{A-a_i} , where $A-a_i = (a_1-a_i, \dots, a_{i-1}-a_i, a_{i+1}-a_i, \dots, a_n-a_i) \in \mathbb{Z}^{d \times (n-1)}$

$$\begin{aligned} \phi_{A-a_i} &:= \varphi_i \circ \Phi_A : (\mathbb{C}^*)^d \rightarrow (\mathbb{C}^*)^{n-1} \\ t &\mapsto (t^{a_1-a_i}, \dots, t^{a_{i-1}-a_i}, t^{a_{i+1}-a_i}, \dots, t^{a_n-a_i}) \end{aligned}$$

$$\varphi_i(X_A \cap U_i) = \overline{\text{im } \phi_{A-a_i}}$$

Prop:

$$\bar{A} = \text{first } r \text{ rows of } Q^{-1}$$

Affine: get a 1-1 parametrization of X_A via Smith normal form (SNF).

Projective: Let $A = 0 \vee (A - a_i) = (0 \ a_2 a_1 \dots a_n a_1) \in \mathbb{Z}^{d \times n}$.

Let $PAQ = S$ be its SNF.

Let $r = \text{rank}(A)$ and let $\bar{A} \in \mathbb{Z}^{r \times n}$ be the matrix consisting of the first r rows of Q^{-1} .

We have $X_A = X_{\bar{A}} = X_{\bar{A}}$ and

parametrization $\bar{\Phi}_{\bar{A}}: (\mathbb{G}^*)^r \rightarrow \text{im } \bar{\Phi}_{\bar{A}}$ of X_A is 1-1.

Prop: minimal affine covering of X_A P

Let $V = \{i: a_i \in A\}$ is a vertex of $\text{Conv}(A)$. We have

$$X_A = \bigcup_{i \in V} (X_A \cap U_i)$$

3.3 The degree of a projective toric variety

Def: degree of $X_A \subset \mathbb{P}^{n-1}$ is the number of intersection points of X_A with a generic linear space of dimension $n-1 - \dim X_A$.

$$P = \text{Conv}(A) \subset \mathbb{R}^d \quad \text{Vol}(P) = \text{Vol}(A) = \int_P dx_1 \dots dx_d.$$

Thm: (Kushnirenko's theorem) If $A \in \mathbb{Z}^{d \times n}$ is such that $\mathbb{Z}'A = \mathbb{Z}^d$, then the degree of $X_A \subset \mathbb{P}^{n-1}$ is the normalized volume $d! \text{Vol}(A)$ of the lattice polytope $P = \text{Conv}(A)$. More generally, if $\text{rank } \mathbb{Z}'A = d$, we have $\deg X_A = [\mathbb{Z}^d: \mathbb{Z}'A]^{-1} \cdot d! \text{Vol}(A)$.

$$\text{lattice index} = |s_1, \dots, s_d|. \quad PAQ = S$$

Thm: Let $X \subset \mathbb{P}^{n-1}$ be an irreducible projective variety of dimension d with coordinate ring $\mathbb{C}[X] = \mathbb{C}[x_1, \dots, x_n] / I(X)$. For $k \gg 0$, the Hilbert function $\text{HF}_X: \mathbb{Z} \rightarrow \mathbb{N} \quad k \mapsto \dim_{\mathbb{C}} \mathbb{C}[X]_k$ is given by Hilbert polynomial $\text{HP}_X(k)$, with leading term $\frac{\deg(X)}{d!} k^d$.

Cor: The Hilbert function of X_A is given by $HF_{X_A}(k) = |kA|$.
 $= \{a_{i_1} + \dots + a_{i_k}, 1 \leq i_1 \leq \dots \leq i_k \leq n\}$

3.4 (Projective) normality and smoothness

Prop: The projective toric variety $X_A \subset \mathbb{P}^{n-1}$ is normal if and only if each of the affine semigroups $IN(A - a_i)$, with $a_i \in \mathcal{V}(\text{Conv}(A))$, is saturated in $\mathbb{Z}'A$.
vertices of $\text{Conv}(A)$

Def: An irreducible projective variety $X \subset \mathbb{P}^{n-1}$ is projective normal if its affine cone $\hat{X} \subset \mathbb{C}^n$ is a normal affine variety.

Prop: Suppose $\mathbb{Z}'A = \mathbb{Z}^d$. $X_A \subset \mathbb{P}^{n-1}$ is projectively normal if and only if the following equivalent conditions are satisfied:

1. $E_{\text{Conv}(A)}(k) = HF_{X_A}(k)$ for $k \in \mathbb{N}$

2. $kA = \{a_{i_1} + \dots + a_{i_k} : 1 \leq i_1 \leq \dots \leq i_k \leq n\} = (k \cdot \text{Conv}(A)) \cap \mathbb{Z}^d$ for $k \in \mathbb{N}$

3. the semigroup $IN\hat{A}$ is saturated in $\mathbb{Z}^{d+1}$.

Prop: Suppose $\mathbb{Z}'A = \mathbb{Z}^d$ and $X_A \subset \mathbb{P}^{n-1}$ is normal. Then X_A is smooth if and only if each of the cones $\text{Cone}(A - a_i)$, with $a_i \in \mathcal{V}(\text{Conv}(A))$, is smooth.

3.5 The projective toric variety of a polytope.

construct a projective toric variety X_P from a lattice polytope P .

Def: Let $P \subset \mathbb{M}_{\mathbb{R}}$ be a lattice polytope and let $P \cap \mathbb{M}$ be the set of its lattice points. P is very ample if the semigroup

$S_P = IN(P \cap \mathbb{M} - v) = IN\{m - v : m \in P \cap \mathbb{M}\}$ is saturated in $\mathbb{M}$ for each vertex $v \in \mathcal{V}(P)$.

Prop: Let $P \subset M_{\mathbb{R}} \cong \mathbb{R}^d$ be a lattice polytope such that $Z'(P \cap M) = M$. Then, P is very ample if and only if the projective toric variety $X_{P \cap M} \subset \mathbb{P}^{|P \cap M| - 1}$ is normal.

Thm: Lattice polytope $\rightarrow$ very ample polytope

Let $P \subset M_{\mathbb{R}} \cong \mathbb{R}^d$ be a d -dimensional lattice polytope. The k -dilation $k \cdot P \subset \mathbb{R}^d$ is very ample for $k \geq d-1$.

Def: Let $P \subset M_{\mathbb{R}} \cong \mathbb{R}^d$ be a lattice polytope of dimension d . Let $k \in \mathbb{N}$ be such that the k -dilation $k \cdot P$ is very ample. The variety X_P is isomorphic to the projective toric variety $X_{(k \cdot P) \cap M} \subset \mathbb{P}^{|(k \cdot P) \cap M| - 1}$.

Prop: "well-defined"

Let $k, l \in \mathbb{N}$ be such that $k \cdot P$ and $l \cdot P$ are very ample. The restriction of $\phi_{k \rightarrow l}$ to $X_{(k \cdot P) \cap M}$ is an isomorphism between $X_{(k \cdot P) \cap M}$ and $X_{(l \cdot P) \cap M}$. Its inverse is the restriction of $\phi_{l \rightarrow k}$ to $X_{(l \cdot P) \cap M}$. 35

Def: A lattice polytope $P \subset M_{\mathbb{R}} \cong \mathbb{R}^d$ is normal if for every $k \in \mathbb{N}$, we have $k \cdot (P \cap M) = \left\{ \sum_{i=1}^k m_i : m_i \in P \cap M \right\} = (k \cdot P) \cap M$.

Cor: Let $P \subset M_{\mathbb{R}} \cong \mathbb{R}^d$ be a d -dimensional lattice polytope. If P is normal, then the projective toric variety $X_{P \cap M}$ is projectively normal.

Thm: Let $P \subset M_{\mathbb{R}} \cong \mathbb{R}^d$ be a d -dimensional lattice polytope. We have the k -dilation $k \cdot P \subset \mathbb{R}^d$ is normal for all $k \geq d-1$.

$X_{k \cdot P \cap M}$ is projectively normal

$\cong X_{P \cap M}$.